

Applied NLP

DATA 641

Lecture 3 — NLP Pipeline; Modeling

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

Modeling

Start with simple heuristics

At the very start of building a model, ML may not play a major role by itself. Part of that could be due to a lack of data, but human-built heuristics can also provide a great start in some ways. Heuristics may already be part of your system, either implicitly or explicitly. For instance, in email spam-classification tasks, we may have a blacklist of domains that are used exclusively to send spam. This information can be used to filter emails from those domains. Similarly, a blacklist of words in an email that denote a high chance of spam could also be used for this classification.

Build your model

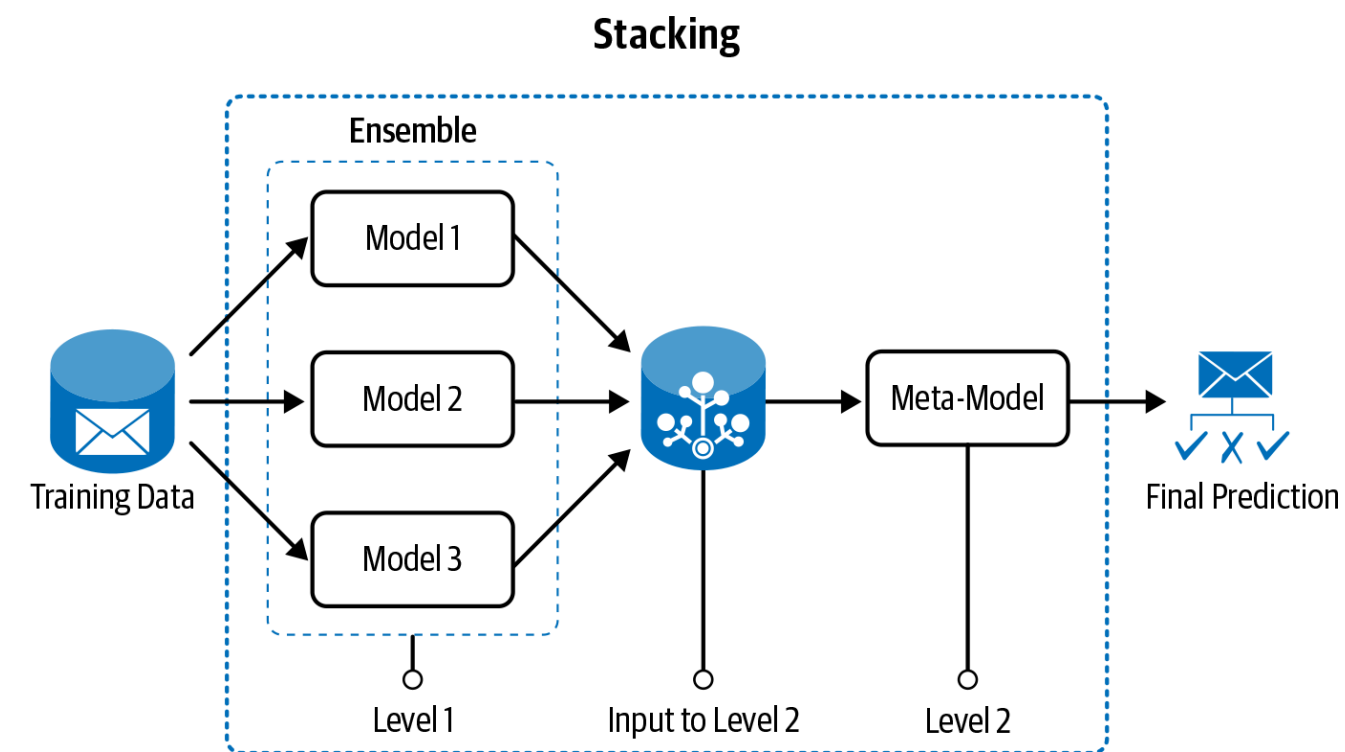
While a set of simple heuristics is a good start, as our system matures, adding newer and newer heuristics may result in a complex, rule-based system. Such a system is hard to manage, and it can be even harder to diagnose the cause of errors. We need a system that's easier to maintain as it matures. Further, as we collect more data, our ML model starts beating pure heuristics. At that point, a common practice is to combine heuristics directly or indirectly with the ML model. There are two broad ways of doing that

- Create a feature from the heuristic for your ML model
- Pre-process your input to the ML model

Building "THE" Model

We have seen examples of getting started building an NLP system by using heuristics or existing APIs, or by building our own ML models. We start with a baseline approach and work toward improving it. We may have to do many iterations of the model building process to "build THE model" that gives good performance and is also production-ready. Some of the approaches to address this issue include:

- Ensemble and stacking



- Better feature engineering
- Transfer learning
- Reapplying heuristics

Data attributes and associated decision paths

Modeling strategies depend heavily on the data at hand. So the following table provides a range of decision paths given our data volume and quality.

Data attribute	Decision path	Examples
Large data volume	Can use techniques that require more data, like DL. Can use a richer set of features as well. If the data is sufficiently large but unlabeled, we can also apply unsupervised techniques.	If we have a lot of reviews and metadata associated with them, we can build a sentiment-analysis tool from scratch.
Small data volume	Need to start with rule-based or traditional ML solutions that are less data hungry. Can also adapt cloud APIs and generate more data with weak supervision. We can also use transfer learning if there's a similar task that has large data.	This often happens at the start of a completely new project.
Data quality is poor and the data is heterogeneous in nature	More data cleaning and pre-processing might be required.	This entails issues like code mixing (different languages being mixed in the same sentence), unconventional language, transliteration, or noise (like social media text).
Data quality is good	Can directly apply off-the-shelf algorithms or cloud APIs more easily.	Legal text or newspapers.
Data consists of full-length documents	Choose the right strategy for breaking the document into lower levels, like paragraphs, sentences, or phrases, depending on the problem.	Document classification, review analysis, etc.